

CSCI 5832

Natural Language Processing

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (63.5 / 100)	41.0 / 100

41

CURRENT AVG SCORE
across 6 job roles

84

POTENTIAL SCORE
with recommended additions

+56

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Explores the field of natural language processing as it is concerned with the theoretical and practical issues that arise in getting computers to perform useful and interesting tasks with natural language. Covers the problems of understanding complex language phenomena and building practical programs.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	63.5	91.3	+27.8	MEDIUM
Machine Learning Engineer	49.1	89.6	+40.5	HIGH
Data Scientist	38.6	83.1	+44.6	HIGH
Applied Scientist	38.0	93.8	+55.8	HIGH
ML Engineer	30.6	72.7	+42.1	HIGH
MLOps Engineer	26.0	75.5	+49.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Business Communication

1 of 6 roles

insights

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

GAP KEYWORDS BY ROLE

AI Engineer

63.5 → 91.3 (+27.8 pts)

perception data models learning infrastructure form software systems

Machine Learning Engineer

49.1 → 89.6 (+40.5 pts)

data learning tools solutions exploratory design development science

Data Scientist

38.6 → 83.1 (+44.6 pts)

toolkit data simply insights social approaches

Applied Scientist

38.0 → 93.8 (+55.8 pts)

lab research models energy drug data learning systems

ML Engineer

30.6 → 72.7 (+42.1 pts)

media data learning systems models infrastructure impact form

MLOps Engineer

26.0 → 75.5 (+49.5 pts)

infrastructure detection data text systems learning media image

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.